Year 1 Evaluation Report for Adventures in Drug Discovery:
Integrating Data Science into the Science Curriculum

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Executive Summary

This external formative evaluation report summarizes the progress and findings of the first year of the Adventures in Drug Discovery (ADD) project, a collaboration between the Institute for Future Intelligence (IFI) and the University of Florida (UF). The project uses the AIMS (Artificial Intelligence for Molecular Sciences) platform developed by IFI to engage teachers with diverse student populations in Next Generation Science Standards (NGSS) practices. Proposed goals are to: 1) Carry out design-based research in collaboration with diverse high schools from six states to develop new educational technologies and curriculum materials to help teachers and students teach and learn data science and AI concepts and skills needed in drug discovery. These materials will be aligned with education standards, integrated into chemistry and biology courses, and improved through iterative design cycles informed by teachers' feedback based on classroom tests. 2) Leverage the complementarity between culturally responsive teaching and inquiry-based learning to develop content that accommodates the unique interests, values, and funds of knowledge of diverse students. For example, new exemplar lessons can draw from historical equity and ethics issues in drug development and show how AI can address these issues through reducing the development cost and targeting more rare diseases. 3) Create a citizen science program for drug discovery based on the products of this project to disseminate them for broader impacts. Such a program will be realized through the gamification of complex computational procedures in drug discovery such as molecular docking and drug screening. This first year of the project initially focused on creating the technological platform and website to support scientific inquiry enabled by data science. It has now turned to preparing curriculum opportunities with a select group of teachers serving as co-designers in the spirit of design-based research.

Expectations for the first year included establishing effective collaboration between IFI and UF; and involving experts in data science, molecular science, and serious games. Overall the evaluation finds the project has exceeded expectations for technology development and innovation and is on track for education innovation and curriculum planning. An area for development involves getting input from expert advisors and partners. Now that the project has

launched with teachers giving input on tools and educational frameworks in the Summer Institute at UF, the project will be making revisions and seeking input from its team of experts.

Evaluation Methods

This first year evaluation is largely a “process evaluation” for formative use by the project team. The emphasis here is on assessing how well the project is ramping up toward a successful longer-term initiative. This includes evaluating the effectiveness of team collaboration, the creation of online tools and educational innovation frameworks and the co-design of curriculum materials and instructional strategies with teachers.

Process Evaluation Questions

The evaluation so far has addressed the following questions:

1. Are activities implemented as intended? If not, what has been changed and why?
2. What characteristics of the project have enabled or hindered reaching its goals?
3. How are materials or activities being made suitable for the intended participants?
4. How well are resources being used – for project development in Year 1?
5. What can be learned about how to create a program like this more smoothly?

Evaluation Methods and Data Sources

Evaluation data sources include meeting logs and minutes (see Appendices), feedback from team members through interviews, and observations of team interactions and collaboration sessions leading up to and including the Summer Institute. In addition to qualitative observations of program activities, we use quantitative analysis of evidence to assess development activities and achievement of goals (e.g., the number of tool features and completed software iteration steps as shown in GitHub’s history).

Findings

Collaborative Development Process

The frequency and quality of team meetings indicate a strong partnership and functioning project. There has been appropriate time between meetings for each team member to pursue independent goals, while the regularly scheduled meetings provide extensive opportunities for feedback and meaning-making. The team meets regularly every two weeks, and this seems to be an appropriate level of frequency.

The first several months saw substantial focus on development and iteration of the online platform and tools that are central to curriculum development and use. During this initial focus on software development (led by IFI) there was a high quality of feedback and questioning about the utility of the tool for engaging in scientific processes and providing worthwhile features for highlighting core curricular concepts.

As the software development reached a more mature stage, the conversations shifted, starting in April (with evaluator encouragement) from tool development to use with educators. At this point the work of UF became pivotal for planning teacher and student engagement, completing the transfer of IRB oversight (completed in June). The team has engaged in

extensive discussions of curriculum goals and teacher co-development processes. Meetings focused on selection of teachers as partners and how to engage them as co-designers, while mapping out incentives and logistics for the first Summer Institute (recently held via Zoom with three teachers on June 19). The project was also represented at the PI meeting in Salt Lake City and used that as an opportunity to engage in collaborative conversations across projects.

Overall collaboration has been strong and collegial with extensive conversations about the technical and scientific challenges being overcome while building a complex learning and teaching platform, and extensive and deep conversations about curricular integration and professional development challenges, particularly given the multidisciplinary nature of the work and learning opportunities. See the Appendices at the end of this report for agendas and materials provided to orient teachers.

Challenges and Solutions

The technology development process has faced and solved numerous technical challenges. These include accessing large data sets related to molecular structures from the PubChem and ChemSpider databases, setting up a system for accurately and easily filtering and screening these for investigation, and creating animations of 3D visualizations based on molecular dynamics simulations.

An administrative challenge involved getting IRB approval for a complex multi-state project that was still being defined in many ways, and doing so within the context of a large university with associated costs and demands. The team had discussions about how to approach this challenge by switching IRB oversight to UF, and the subsequent IRB application was approved by UF with arrangements made to include the external evaluation.

A curricular integration challenge involved achieving goals of diversity, equity and inclusion (DEI), given the current political situation in Florida. With legislation prohibiting this as an explicit focus, the team discussed reaching “all learners” in the Florida schools and expanding participation to underrepresented populations outside of Florida (e.g., in Hawaii). The project will continue working with teachers with diverse students, with an emphasis on public school classrooms, and the evaluation will continue to monitor diversity among students who are selected for participation.

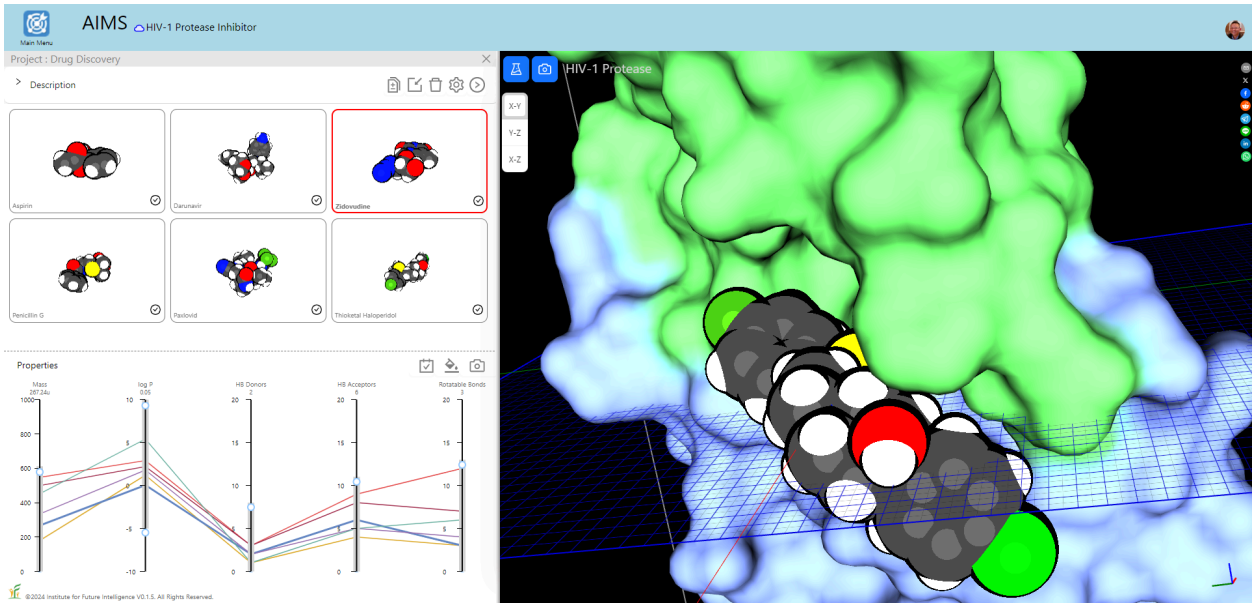
A final challenge concerned timing the work to balance the desire for feedback from teachers and the experts with the need for having something concrete to show. The project has now created the technological tools and provided a list of curriculum opportunities for discussion with teachers so it is ready to seek additional input and feedback from supporting institutions and experts. This has seemed like an appropriate use of resources and a way to avoid “too many cooks” while also ensuring the experts have something concrete and actionable to review. This mirrors the decision to involve teachers and discuss curriculum opportunities after the tools were available to demonstrate the capabilities that were being offered

Focus # 1: Technology Innovation: The AIMS Platform

The first focus for the project has been online tool development, allowing for the creation of curriculum materials powered by visualization of complex computational procedures in drug discovery. Significant progress has been made, as evidenced by demonstrations in ongoing

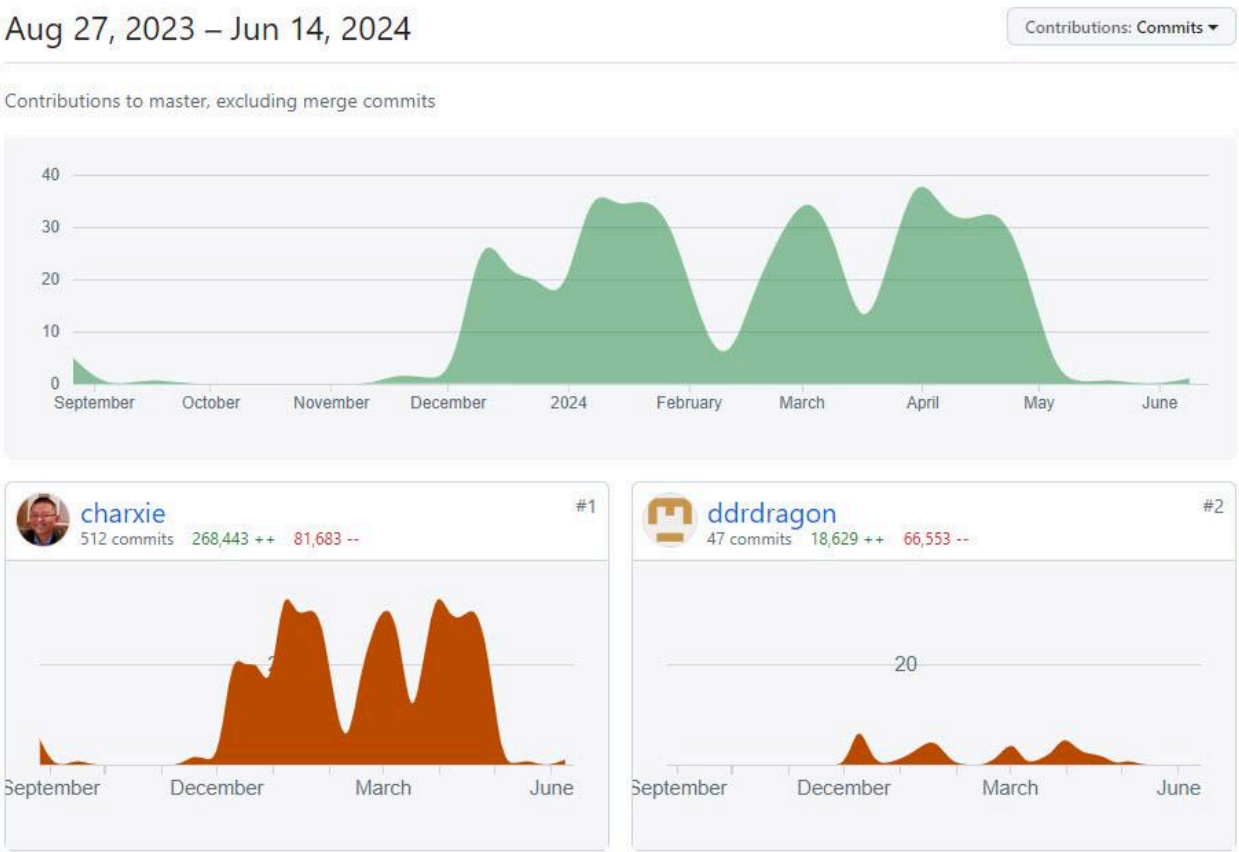
team conversations and witnessing the growing set of functionalities and topics available through the AIMS website. Technology development discussions included building the interactive single-page app based on React, and use of visual analytics such as screening and filters to narrow down the number of candidate molecules under investigation and being visualized. Another innovation being planned is the use of gamification to engage students in a scientific area that is typically considered as abstruse.

Specific technology accomplishments include importing the large data set, filtering and screening selection of molecules, creation of the web interface and integration of data science tools. In the case of drug discovery, AIMS features a ligand galley (on the left in the image below), and the docking chamber with a target protein (on the right).



AIMS can import a ligand into the chamber to conduct docking tests with the selected protein. It allows the user to select candidate molecules and target proteins from a preset library, housed in an internal database curated from PubChem/ChemSpider and Protein Data Bank (PDB). Users can save any such project in their personal gmail-based cloud account in AIMS. This permits users to have as many projects as they want, each targeting a different problem.

Overall, the project has had success in building a highly interactive Web interface for visualization of molecular dynamics and molecular docking that will be used for educational and drug discovery purposes. Evidence of technology development is visible on the project website, which now demonstrates advanced capabilities, many of which were only available in professional cheminformatics software. The evidence of technology development can be shown in the GitHub logs below.



Another technical milestone involves the mechanism for visualizing molecular docking, which is still under development. The AIMS tool is set up to offer engaging 3D visualization on the right side of the interface showing the interactions of selected molecules in a way that resembles a docking-station / tetris-type game.

The newly developed and innovative platform can now help students make sense of the results from machine learning directly using a relatively simple user interface. Users can select training and test data sets using this tool. Going beyond expectations, the project no longer needs to use an external tool (such as iFlow) to do this. This groundbreaking tool offers functionalities needed to implement the project, as discussed in the alkane unit, the first example from the original ADD proposal. All the browser-based technology pieces are now fully functional as shown below and online here: https://intofuture.org/aims-alkanes.html.

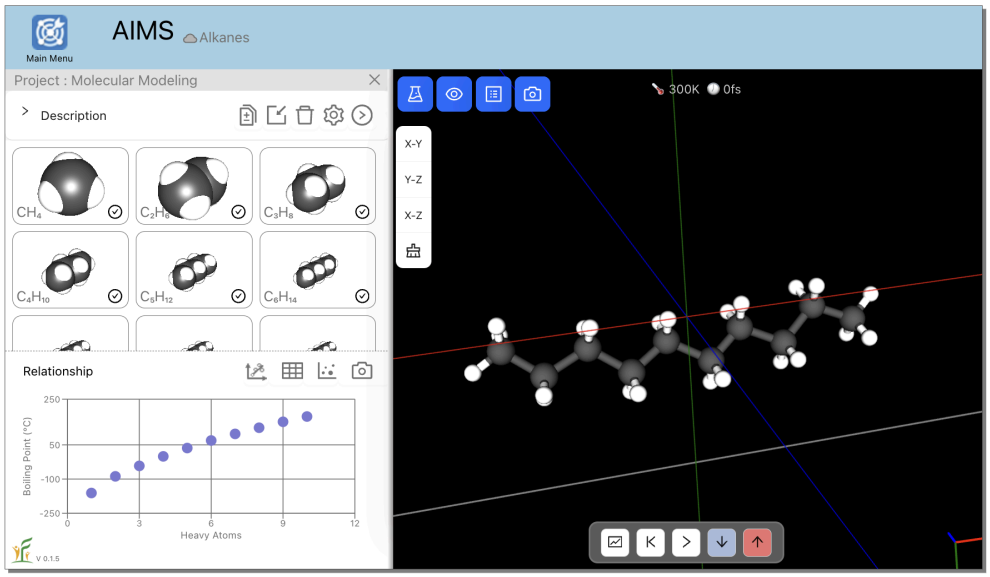
In summary, the technological innovation so far is impressive. The project plans to build more tools such as Artificial Neural Networks (ANN) to provide more options for Quantitative structure-activity relationship (QSAR) modeling [1], with an emphasis on user interface (UI) work, based upon co-design feedback from teachers and students, to support these tools. The additional screenshots, also shown below, highlight the project's scientific and technical innovations related to drug discovery and molecular modeling.

Quantitative Structure-Property Relationships of Linear Alkanes

By Charles Xie

Back to AIMS home page

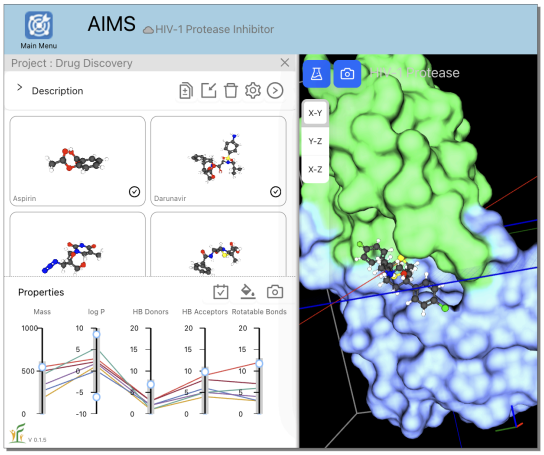
Linear alkanes consist of hydrogen and carbon atoms with a general chemical formula C_nH_{2n+2}. In the following window, the gallery on the left shows a number of alkanes with increasing chain length and the simulation on the right brings alkanes to "life" on your screen — You can observe how an alkane molecule moves (including different modes of translation, rotation, and vibration).



Live model above (view in full screen) — Chrome or Edge recommended

Drug Discovery

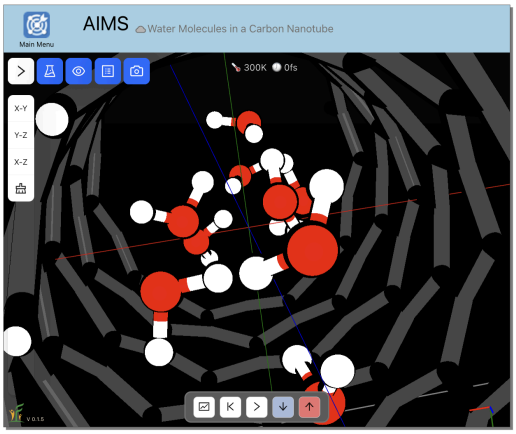
Small molecules represent an important type of drugs — 90% of all drugs sold on the global market are small molecules. The following example shows the process of screening an inhibitor to an HIV protease from a collection of small molecules using AIMS. Interactive molecular docking, currently under development, will allow users to validate a candidate molecule selected based on data science principles. For example, Lipinski's Rule of Five is implemented in AIMS as filters in the hyperspace of molecular properties as shown below.



Live model above (view in full screen) — Chrome or Edge recommended

Molecular Modeling

Interactive molecular dynamics allows users to design and perform computational experiments to test their own hypotheses. Play with a molecular dynamics simulation below to experience the power of computation in the browser!



Live model above (view in full screen) — Chrome or Edge recommended

With the visual analytics supported by techniques such as brushing and linking built in AIMS, users can explore quantitative structure–activity/property relationships (QSAR/QSPR) of molecules that they have collected. A well-known QSPR is the relationship between the boiling points and the carbon atom counts of alkanes (see this case study based on AIMS).

Source: https://intofuture.org/aims.html

Focus #2: Educational Innovation and Opportunities for Teaching and Learning

The above scientific and technical accomplishments must be matched to educational opportunities. With the above tools having been built, the focus has shifted to identifying powerful educational uses and engaging and recruiting teachers for co-design. Expectations for this part of the project involve 1) Development of new educational technologies and curriculum materials aligned with education standards; 2) Integration of these materials into chemistry and biology courses, and 3) Iterative improvement of materials based on teacher feedback.

For the last several months the team has been focusing intensely on solving curriculum and professional development (PD) challenges. This includes addressing standard concerns related to recruiting the best possible teachers to work as initial co-designers to test these tools, offering appropriate incentives for participation, and placing these within course content and educational standards. Recruiting of teachers has focused on subject-area teachers with varied areas of expertise involving chemistry literacy, biology literacy, data literacy for STEM, and experience with creating culturally responsive curriculum. The project was successful in getting the IRB approved and running a Summer Institute to build out a curriculum with a few of these teachers, with plans for systematically adding participants in the coming years.

A key challenge has been identifying the nature of questions to be addressed by students and teachers given the range of topics that can be addressed across multiple domains of chemistry, biology and data science. The project has extensively discussed the need to define the scope of the “curriculum unit” and avoid doing too much – while still accommodating multiple disciplines – and avoiding overwhelming the minds of the learners. An initial list of core content from NGSS that can be taught using the tool was identified and shared with teachers leading up to the Summer Institute (see Appendices). These include the following:

Physical Sciences
PS1: Matter and its interactions
PS2: Motion and stability: Forces and interactions
PS3: Energy
PS4: Waves and their applications in technologies for information transfer

Life Sciences
LS1: From molecules to organisms: Structures and processes
LS2: Ecosystems: Interactions, energy, and dynamics
LS3: Heredity: Inheritance and variation of traits
LS4: Biological evolution: Unity and diversity

Engineering, Technology, and Applications of Science
ETS1: Engineering design
ETS2: Links among engineering, technology, science, and society

Chemistry content learning resources are also available on the AIMS website, with vignettes for using this tool to explore various chemistry concepts: Molecular dynamics, Ionic bonding, Covalent bonding, Diatomic molecules, Triatomic molecules, Linear alkanes, Cycloalkanes, Acenes, Cis-trans isomerism, Buckyballs, Water in a nanotube, Face-centered cubic lattice, Body-centered cubic lattice, Diamond vs. graphite, Graphenes, Zeolites, Protein alpha helix, DNA double helix and Coarse-grained models.

Pedagogical conversations emphasized strategies for getting students (and teachers) thinking about these topics and the overlap among data science, chemistry and biology. The team discussed and adopted the use of Understanding by Design principles and project-based learning ideas for creating a “need to know” content to solve real-world problems. The team has discussed using “anchors” for instructional scaffolding to guide attention and avoid cognitive overload. The project is also aware of and seeks strategies to address the potential problem of students “playing around” with the molecules without promoting understanding.

To engage all learners from diverse communities (including students under-represented in STEM) the project has explored (with evaluator input) a motivational framework based on Keller (ARCS). This includes drawing attention (A) to key issues, establishing relevance (R), creating a feeling of confidence (C) and satisfaction (S). This aligns with the team’s development of a focus on drug discovery activities for treating diseases with local and cultural relevance (e.g., sickle cell for African Americans), and other neglected tropical diseases (NTDs) that originate and are most prevalent in tropical regions. To draw attention and frame relevance the project discussed the need to address NTDs from the World Health Organization, and the effect on poor people and people of color in the US and elsewhere. Additional diseases of interest include dengue and malaria, which have been addressed using computational tools in ways that parallel the functionality of AIMS [2].

Future curricular materials development includes plans for embedded assessments and with more explicit connections to education standards and measures (see proposal). Additional educational tools to be developed include a “Quick-start guide” and updates to the full user manual that has already been created so others can follow. These will provide well-developed example applications that serve as a demonstration for others to explore along with a student workbook and a teacher guide. This will take time and is going to be built while engaging the teacher co-designers to ensure it has practical applications.

Finally, for education outcomes, the project has clarified its intention to give students ideas and to develop their confidence in how to solve problems, and create interest in professional opportunities based on the satisfaction from how it feels to “help cure a disease”. The team discussed the motivational aspects of addressing authentic problems (similar to CS outreach programs), with different types of authenticity to explore including personal meaningfulness and health authenticity, familial and community health authenticity, societal connections, as well as scientific authenticity, and benefits due to job authenticity. In short, ADD seeks to convey and assess (and the evaluation will explore) the belief that drug discovery matters for students and matters for society, while building student career interests and self-efficacy (e.g., for eventually working for pharmaceutical companies or public health sectors).

Summer Institute (Year 1)

The Summer Institute, launched on June 19, 2024, was intended to initiate the larger focus on co-design and to make sure that a small number of example projects would be developed that illustrate strong pedagogy and content focus as outlined above. These example projects will be tested with students and then used in future years as starting points for some new teacher participants to use in a customization process for their unique contexts. Year 1 participants included three teachers, previously known by the UF team, with incentives for participation offered as detailed in the project proposal. While the emphasis has been on biology and chemistry, and NGSS alignment, there was also consideration to how the project can contribute to an understanding of data science, use of data to accelerate scientific discovery, and how to view “molecules as data”.

The agenda introduced the project and design concepts, reviewing the user interface to illustrate how the UI works and core operability. This part was intended to answer questions about AIMS (the tool) – What does it do? What can I do with it? How do I do it? This included explanations about the functionality of the tool and the use of filters to identify key characteristics of different molecules and to afford an understanding of molecules as the vocabulary for working with AIMS. The Institute presented AIM’s use of QSPR methods as an example of the connection among Chemistry/Biology content ideas, science and computational thinking practices, and structure-function relationships as a cross-cutting concept. Structure-function relationships in chemistry are commonly associated with boiling point, melting point, and toxicity, but are also applicable to drug discovery.

Summer Institute Feedback Survey

Evaluation of the Summer Institute included observations of interactions and questions, exploration of tool use and questions or challenges that were observed, and a short post-Institute survey conducted independently by the evaluator via email and an online form. (see Appendices). The following summary of the survey responses is consistent with what was observed by the evaluator during the session.

Responses indicate that the perceived purpose was multifaceted, including to introduce the AIMs interface and its potential uses, clarify expectations and deliverables, and provide an overview of the modeling systems while facilitating team member introductions. The most valuable aspects according to the participating teachers involved learning how to navigate and begin a project in the AIMs interface, understanding its limitations and in-depth features, and beginning to see potential classroom applications of the modeling program. The expert walkthrough of the platform was particularly beneficial for understanding the interface and locating specific features.

The responses suggest several improvements for future sessions to enhance their helpfulness and productivity: 1) Providing a detailed layout of the project being demonstrated would help participants follow along more effectively and avoid confusion with their data. One wrote “The demonstration started off well, and then at a certain point I was having a hard time following along, and ended up with data that looked much different than the one being shown”. 2) Attendees should come prepared with specific questions related to their lesson goals to make the sessions more relevant and beneficial. One wrote “making sure I attend with specific

questions tailored to my lesson goals will definitely make future sessions more helpful and productive." Finally, 3) Having more focused discussions. While debates on parameters and limitations of the modeling system were acknowledged as important, feedback suggests they should be more aligned with the practical needs of high school classrooms to ensure they contribute effectively to curriculum development. As one teacher wrote "The debate is beyond the content of a high school classroom."

Quantitative responses used a rating scale of 1-lowest to 10-highest. There were no responses below 6, which suggested that overall the session received high ratings. The largest range of responses was from 6 to 9. The responses from low to high for each item were as follows, with the lowest mean being 7.33 regarding the clarity of the information and a high mean of 9.0 for feeling views were listened to and taken seriously

1) How clear was the information that was provided?
   7, 7, 8; Mean = 7.33

2) How well do you feel your views were listened to and taken seriously?
   8, 9, 10; Mean = 9.00

3) How excited are you about using the AIMS in the classroom?
   8, 8, 10; Mean = 8.67

4) How confident do you feel about taking the necessary next steps to use AIMS in the classroom?
   6, 8, 9; Mean = 7.67

In clarifying their numeric responses, two of the teachers expressed that while the session began clearly and concisely, it became difficult to follow, particularly during an in-depth discussion on linear versus polynomial topics, which overshadowed the demonstration of the AIMs interface. Additionally, one teacher expressed lingering uncertainty, that somewhat diminished enthusiasm, about how AIMs will be implemented in the classroom. Overall these added comments indicate that teachers will benefit from clearer, more practical demonstrations and focus on classroom applications.

The teachers identified several necessary next steps for effectively using AIMs in their classes: 1) Aligning with Standards: Reviewing educational standards and selecting appropriate ones to design a unit, especially considering state-specific standards; 2) Gaining knowledge about the process and limitations of importing new molecules into AIMs to know what can be requested. 3) Learning about specific dataset parameters and their measurements, such as how hydrogen bond donors and acceptors are measured in the PubChem dataset and 4) Ensuring the AIMs system is approved by their district regarding compliance with student privacy requirements. This included a request for more information in the future to address data sharing and student privacy controls to ensure compliance and security.

Other topics teachers would like to see addressed included alternative focus areas, specifically exploring topics beyond alkanes, such as neglected tropical diseases, to

demonstrate the benefits of AIMs in biology classrooms and providing more explicit visions from developers on how AIMs approximates machine learning and AI, and how it can be integrated into classroom instruction.

Overall Conclusion

The process evaluation for the first year of the ADD project indicates that the project is on track toward achieving its long-term goals. Critical components of this success that appear to be in place include effective team collaboration, the creation of technically sophisticated but usable and engaging online tools, and teacher involvement in the development of curriculum materials that can be mapped to core content standards. The evaluation will continue monitoring for evidence of refinements that will be essential to maintain this trajectory.

Recommendations

Some recommendations based on work so far include

- Foster continued communication and collaboration among team members and external experts.
- Continue iterative improvements of curriculum materials based on teacher feedback (outlined above) and, eventually, student feedback
- Initiate pilot testing of online tools to gather comprehensive data on user engagement and effectiveness.
- Incorporate the “drug discovery cycle” to scaffold relevant learning tied to authentic drug discovery processes. (The proposal illustrates an iterative process for investigating a disease, selecting target receptors, going through the process of investigation and communicating results, etc.)

Lessons Learned

The project has learned lessons that highlight

- The importance of regular and structured team meetings to ensure effective collaboration.
- The value of listening to teacher feedback and engaging in iterative feedback loops in developing high-quality educational materials with teachers
- The potential of gamification to enhance engagement in citizen science programs.

Next Steps for the Project

In the coming years, the project and evaluation expect to produce findings on the effectiveness of the online tools and findings on the engagement and participation of learners. The project will document interactions and use by students and their reactions to the technology and pedagogical opportunities. Feedback from pilot testing will include documentation of user interactions with AIMS and larger surveys of users as the project scales. We expect to see the following activities

- Continue iterative improvements based on teacher and student feedback.
- Expanding the reach of the citizen science program to engage more participants.

- Published findings in peer-reviewed journals to inform the broader STEM education community.

We expect to learn lessons about the importance of culturally responsive teaching in engaging diverse student populations based on the selection of project activities related to drug discovery (e.g., NTDs) and the potential of gamification to enhance participation in citizen science programs related to molecular chemistry, biology, data science and drug discovery.

Next Steps for the Evaluation

Future evaluation reports expect to use a mixed-methods approach, including surveys, pre/post-assessments and learning analytics (from IFI), and expert reviews of curriculum materials (e.g., using the NGSS's EQuIP Rubric for Science). Additional data sources are expected to include student and teacher questionnaires, process data from user interactions with AIMS, and teacher interviews. These data will help address the following questions:

- For students: How effectively does the integration approach a) improve science learning towards education standards, b) provide opportunities for learning to address diversity, equity, and inclusion and c) increase student interest in biomedical and/or computing careers?
- For teachers: How effectively does the integration approach a) help chemistry and biology teachers teach the core content in their classes, b) engage teachers in adopting culturally responsive curriculum and instruction, and c) invite science teachers to include data science in their teaching?
- For broader impacts: How effectively does the project a) produce materials used by a growing number of teachers and students?, b) disseminate a gamified version of AIMS to engage the public, c) promote culturally meaningful citizen science in diverse communities, and d) result in peer-reviewed publications to inform the STEM education communities?

Limitations

Limited data from the first year may not fully capture the long-term impact of the project. Year 2 will address challenges in rolling out curriculum pilots and collecting comprehensive data from all participating schools, teachers and students.

To conclude, this evaluation report has provided a structured overview of the first year's progress and sets the stage for continued project work and evaluation in subsequent years. The first year of the ADD project has made significant progress towards its specific aims. The development of educational technologies and curriculum materials, the creation of culturally responsive learning opportunities and content, and the establishment of citizen science activities are on track. Further data collection and analysis as work proceeds will provide more detailed insights into the project's impact in the coming years.

REFERENCES

[1] Sato A, Miyao T, Jasial S, Funatsu K. Comparing predictive ability of QSAR/QSPR models using 2D and 3D molecular representations. J Comput Aided Mol Des. 2021 Feb;35(2):179-193. doi: 10.1007/s10822-020-00361-7. Epub 2021 Jan 4. PMID: 33392949.

[2] Ncube, N.B., Tukulula, M. & Govender, K.G. Leveraging computational tools to combat malaria: assessment and development of new therapeutics. J Cheminform 16, 50 (2024). https://doi.org/10.1186/s13321-024-00842-z

APPENDICES

Appendix A. TEACHER PREPARATION MATERIALS FOR SUMMER INSTITUTE

Agenda

- Zoom Link
- Introductions (10min; Kent)
- Educational technology: How AIMS supports NGSS 3D learning (1hr.; Charles)
- Discussion of Initial Curriculum Activity Ideas (30min; Kent)
- Logistics: W-9 form, consultant service agreement, invoice, payment (10min; Charles)

Friday, May 31st, 2024

Agenda

- Getting the Process Started
- Orienting to the project and the goals (slides)
- Things to start learning more about **before the kickoff meeting** (you don't need to know everything right away, our model is based on gradually learning more over time, it's just time to start)
    - ADD (goals and purpose of the project)(pdf)
    - AIMS Features (concepts and ideas that shape the intent of the software)
    - Three Dimensional Learning (the focus of NGSS)
    - Neglected Tropical Diseases (NTDs)
    - Understanding by Design (UbD) (pdf)
    - Culturally-Relevant Curriculum and Instruction

Action Items

Agenda

- Introductions
- Discuss Project Overview (KC)
- Q&A to negotiate details (All)

Attendance (CH, DO, KC, WF)

FYI Items

- ADD Curriculum Design Description.pdf
- ADD Background-Purpose.pdf

Getting the Process Started

Directions: Please read through the linked resources in each section and use the information in the readings to answer each question in 3 to 5 sentences. Include your responses by editing this document.

Section 1 : The Perspective and Goals of ADDS

Background Reading:

- ADD Background-Purpose
- ADD Curriculum Design Description

1. Based upon your initial reaction, what standards can you imagine aligning your unit with? (*See p.2 (below) for a list of possibilities. Feel free to suggest others not contained within the list.*)
2. The ADDS Background-Purpose document includes the following statement "How can the three-dimensional learning of NGSS be implemented with the support of data science as the "fourth pillar" of science?" What initial ideas do you have for this question?
3. At what time(s) during the year (i.e., months) could you envision your lesson fitting into the content curriculum?

Section 2 : (NTDs as Culturally Relevant Issues for Curriculum Design)

Background Reading:

- Neglected Tropical Diseases

1. In what ways can you imagine NTDs serving as the context and or content of your curriculum? (including ways that they might already be used)
2. Looking at your roster for next year, how could the NTD that you identified in question 1 be construed as culturally relevant curriculum for your students and why?
3. What other ideas do you have, if any, about using AIMS in culturally relevant ways (including alternatives to NTDs)? See for example: A Culturally Relevant Collab with NGSS and Chemistry Chem Ed Xchange.pdf

Section 3 : UbD as a Curriculum Development Process

Background Reading:

- Using Understanding by Design to Make the Standards Come Alive

1. Focusing on Stage 1 of Backwards Design (Identify desired results), what do you imagine could be the ultimate goal of your unit?
    a. What should your students **know**?
    b. What should your students **understand**?
    c. What should your students be able **to do**?

Possible Standards/Concepts

Disciplinary Core Ideas
- PS1.A
- LS1.A

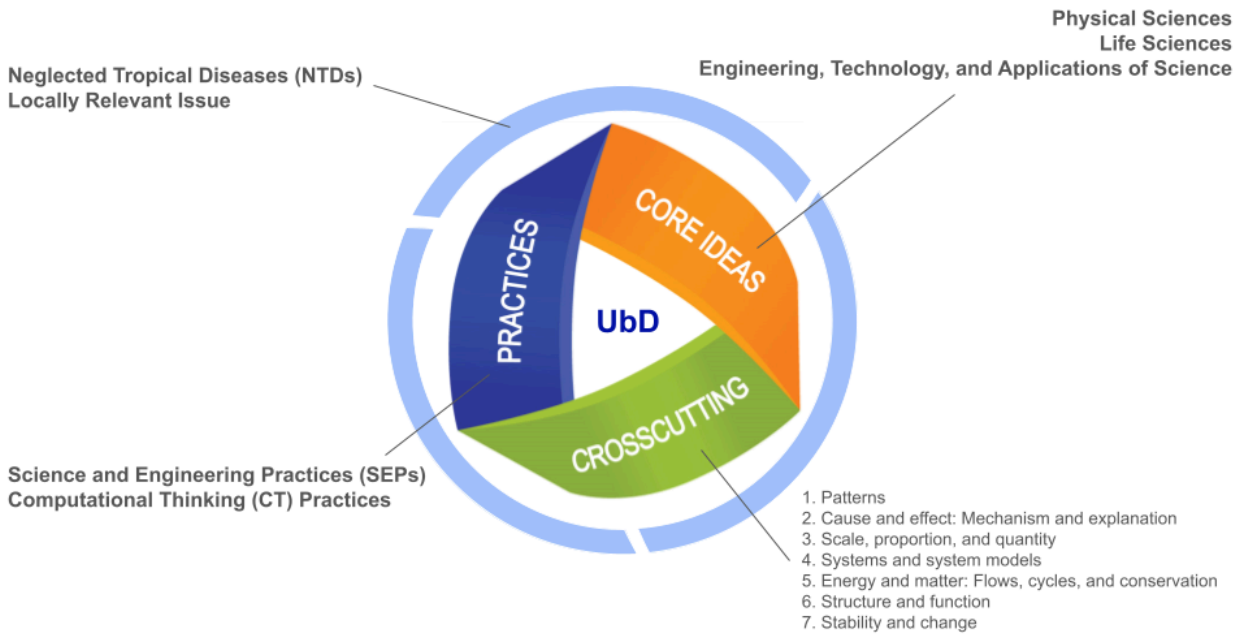
Cheminformatics
- quantitative structure-property/activity relationship (QSPR/QSAR)

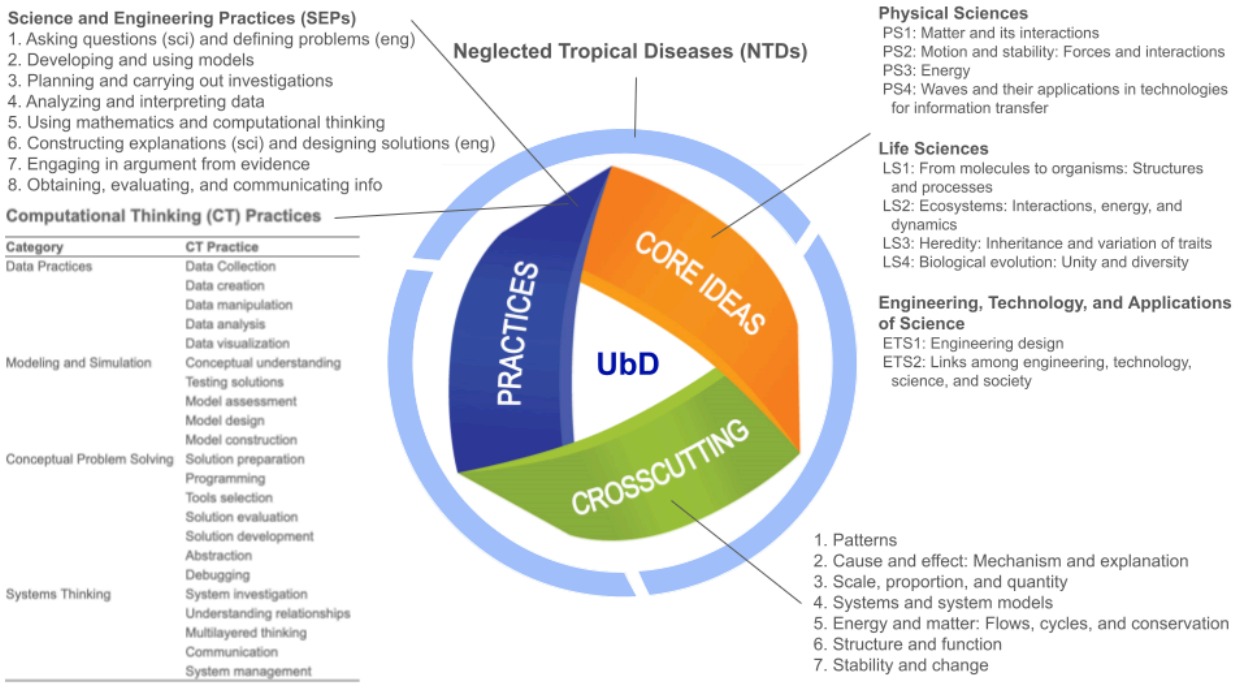
Cross Cutting Concepts
- Patterns
- Cause and Effect
- Structure and Function

Science and Engineering Practices
Option of design challenge?
1. Ask a question related to a chemical property or biological activity
2. Collect information about relevant molecules from public databases to prepare a training dataset
3. Find patterns in the training dataset and build a mathematical model to approximately represent them
4. Use the model to predict the properties of other molecules
5. Validate the results with a test dataset
6. Repeat steps 2-5 to refine the model as needed.

K12 Computer Science Standards
Operationalizing Data Science
- Ask the right questions.
- Explore and collect data 3A-DA-10, 3B-DA-06
- Extract and clean data 2-DA-08 4
- Handle missing values and outliers
- Visualize and analyze data 1B-DA-06, 3A-DA-11
- Find patterns in data 2-DA-09, 3B-DA-05
- Make predictions with models 3A-DA-12
- Present the results 1B-DA-07, 3B-DA-06/07

Selected Slides for Summer Institute on Curriculum Co-Design





Category	CT Practice
Data Practices	Data Collection
	Data creation
	Data manipulation
	Data analysis
	Data visualization
Modeling and Simulation	Conceptual understanding
	Testing solutions
	Model assessment
	Model design
	Model construction
Conceptual Problem Solving	Solution preparation
	Programming
	Tools selection
	Solution evaluation
	Solution development
	Abstraction
	Debugging
Systems Thinking	System investigation
	Understanding relationships
	Multilayered thinking
	Communication
	System management

Post Summer Institute Evaluation Survey (June 20, 2024)

Adventures in Drug Discovery (ADD) 2024 Summer Institute Evaluation Survey

In your own words, what do you believe the purpose of today's session was? *

Your answer

What was the most valuable aspect of today's session for you? *

Your answer

What could be improved to make future sessions more helpful or productive? *

Your answer

How clear was the information that was provided? *

	1	2	3	4	5	6	7	8	9	10	
Lowest rating	○	○	○	○	○	○	○	○	○	○	Highest rating

How well do you feel your views were listened to and taken seriously? *

	1	2	3	4	5	6	7	8	9	10	
Lowest rating	○	○	○	○	○	○	○	○	○	○	Highest rating

How excited are you about using the AIMS in the classroom? *

	1	2	3	4	5	6	7	8	9	10	
Lowest rating	○	○	○	○	○	○	○	○	○	○	Highest rating

How confident do you feel about taking the necessary next steps to use AIMS in the classroom? *

	1	2	3	4	5	6	7	8	9	10	
Lowest rating	○	○	○	○	○	○	○	○	○	○	Highest rating

(optional) If necessary, please clarify the above responses

Your answer

What are some necessary next steps to allow effective use of AIMS in your classes?

Your answer

Are there any other topics would you like to see addressed?

Your answer

(optional) Do you have any other feedback or comments?

Your answer

(optional) To volunteer for a 10 minute follow-up interview please provide your email here.

Your answer